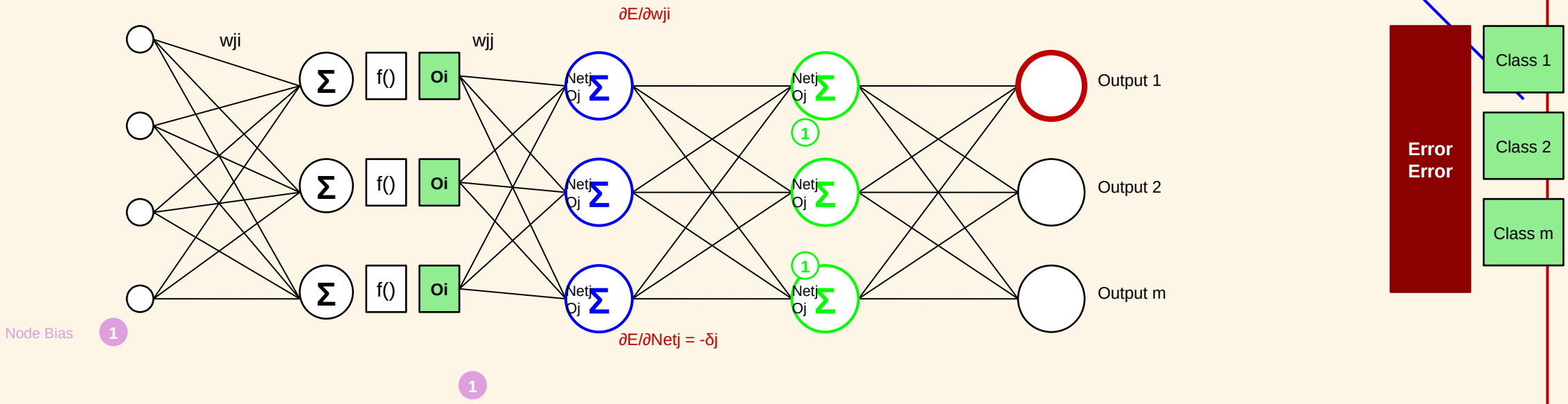
Updated Weights of Output Layer

(Calculated Error is affected by Netj,)

Input Sample

Target Label



$$\Delta_{ji} = -\eta \frac{\partial E}{\partial w_{ji}} = -\frac{\partial E}{\partial Net_j}\frac{\partial Net_j}{\partial w_{ji}} = \eta \delta_j \cdot O_i$$

Where δj = -∂E/∂Netj (one δ for each Node j)

Layer, Sigmoid Activation $\Rightarrow \delta_j = (t_j - O_j) \cdot O_j(1 - O_j) \Rightarrow \Delta_{ji} = \eta(t_j - O_j) \cdot O_j(1 - O_j) \cdot O_i$